

Greatest common factor



1

a 3

b a

c $3x$

d $3b$

2

a $4r + 12 = 4(r + 3)$

b $5v - 30 = 5(v - 6)$

c $18c - 24 = 6(3c - 4)$

d $2t^2 + 2t = 2t(t + 1)$

3

a $6(v + 5)$

b $5(9t - 8)$

c $-2(s + 5)$

d $-2(6s - 5)$

e $y(y + 4)$

f $k(k - 5)$

g $2u(u - 4)$

h $2t(2 + t)$

i $x(42 - x)$

j $7(x + 5)$

k $6(6 - x)$

l $-2(x - 5)$

m $-10(10 + x)$

n $-6(3y^2 - 1)$

o $13(3x + 4)$

p $(t - 3)(8t + 9)$

Grouping with 4 terms

4

a $(5 + v)(a + b)$

b $(y - z)(x - w)$

c $(2y + 3w)(8x - z)$

d $(5x^2 + 4y)(8z - 1)$

e $2(y - 4)(4y - 5)$

f $(4c - d)(c + 9d + 5)$

5

a No

b Yes

c Yes

d No

6

a $(x + 5)(x + 8)$

b $(x - 5)(x + 10)$

7

a $(x + 2)(y - 9)$

b $(8 - x)(7 - y)$

c $(x + z)(12y + w)$

d $(x - 20y)(2 + z)$

e $(5 + x)(10 + y)$

f $(12y + w)(x + z)$

8 $(x + 4)(4x + 1)$



Trinomials

9

a $m + 10$

b $y + 5$

10

a $(x + 1)(x + 5)$

b $(x + 4)(x + 10)$

c $(x - 1)(x - 4)$

d $(x - 3)(x - 10)$

e $(x - 8)(x - 7)$

f $(x + 12)(x - 11)$

g $(x + 15)(x - 6)$

h $(x + 2)(x - 4)$

i $(x + 7)(x - 10)$

j $(x + 2)(x + 3)$

k $(x - 11)(x - 4)$

l $-(x + 1)(x + 12)$

m $-(x - 9)(x - 1)$

11

a $4(x + 6)(x + 8)$

b $-5(x - 4)(x + 2)$

c $3(x - 9)(x + 2)$

d $2x(x + 5)(x + 3)$

12

a $3x^2 + 10x - 8 = (3x - 2)(x + 4)$

b $12x^2 + 25x + 7 = (4x + 7)(3x + 1)$

c $15x^2 - 13x - 6 = (5x - 6)(3x + 1)$

13

a $(7x - 5)(x - 10)$

b $(6x + 7)(x - 1)$

c $(9x - 10)(x - 1)$

d $(7x + 8)(x - 6)$

e $(3x + 2)(2x + 3)$

f $-(4x - 3)(3x + 4)$

g $(9 - 8x)(10 - x)$ or $(8x - 9)(x - 10)$

h $-(7p - 2)(7p + 4)$

14

a $(b + 7)^2$

b $(y - 5)^2$

c $(x + 5)^2$

d $(x + 7)^2$

e $(x - 6)^2$

f $(x - 8)^2$

g $(9x - 4)^2$

h $(9x + 2)^2$

i $(7x - 2)^2$

j $(9 + x)^2$

k $(6 - x)^2$

l $(5 + 6r)^2$

m $7(d + 8)^2$

n $4(x + 5)^2$

o $3(x - 4y)^2$

p $-6(r + 8)^2$

q $-3(x - 2)^2$

r $\left(b - \frac{7}{2}\right)^2$

s $\left(x + \frac{3}{2}\right)^2$

t $\left(x - \frac{3}{2}\right)^2$



Difference of squares

15

a $p + 8$

b $m + 6$

16 $x - 7$

17 $(x^2 - 10)(x^2 + 10)$

18 $(ax - b)(ax + b)$

19

a $(x + 5)(x - 5)$

b $(n - 6)(n + 6)$

c $(v - 7)(v + 7)$

d $\left(x - \frac{2}{7}\right)\left(x + \frac{2}{7}\right)$

e $(9 - x)(9 + x)$

f $(2 - u)(2 + u)$

g $(2 + 7y)(2 - 7y)$

h $(8x - 11)(8x + 11)$

i $(11m + 8)(11m - 8)$

j $(9x - 10y)(9x + 10y)$

k $(11x + 7y)(11x - 7y)$

l $(4x - 5)(4x + 5)$

m $6(x - 7)(x + 7)$

n $3(t + 2)(t - 2)$

o $5(x - 8)(x + 8)$

p $4(n + 2)(n - 2)$